Supply vs. Demand Shocks and Distributional Inflation: Do Low-Income Households Bear Greater Burden When Shock Origins Differ?

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Abstract

This paper investigates whether the distributional consequences of inflation differ systematically depending on whether inflation is driven by supply or demand shocks. Using the San Francisco Federal Reserve's supply/demand PCE decomposition (Shapiro 2022), income-stratified CPI indices from BLS Consumer Expenditure Survey budget shares, and VAR impulse response analysis over January 2000 through March 2026, I find that supply shocks are approximately **1.88 times more regressive** than demand shocks. During supply-dominant periods, the bottom quintile faces an inflation premium of 0.58 ppts over the top quintile; during demand-dominant periods this collapses to 0.025 ppts ($t = -6.32$, $p < 0.001$). A post-COVID structural break is confirmed (Chow $F = 127.45$, $p < 0.001$), with the 2021–2023 period exhibiting the largest sustained inflation gap at +0.81 ppts.

Keywords: distributional inflation, supply shocks, demand shocks, income inequality, VAR

JEL: E31, E52, D31, I32

1. Introduction

Inflation is not experienced uniformly. Low-income households allocate large shares of their budgets to necessities while high-income households spend more on discretionary goods. A Federal Reserve Board (2025) working paper establishes that adverse shocks systematically increase low-income household inflation through non-homothetic demand. This paper asks whether the distributional burden differs depending on whether inflation is supply- or demand-driven — a distinction with direct implications for monetary and fiscal policy responses.

I find: (1) supply shocks are 1.88× more regressive than demand shocks (coefficients 0.613 vs. 0.326); (2) the inflation gap is near-zero in demand-dominant periods (0.025 ppts) versus substantial in supply-dominant periods (0.579 ppts; $p < 0.001$); (3) a structural break at COVID-19 is confirmed (Chow F = 127.45), with the largest sustained gap in the 25-year sample (+0.806 ppts) during 2021–2023.

2. Literature Review

2.1 Distributional Inflation

Hobijn and Lagakos (2003) document cross-household inflation heterogeneity. Argente and Lee (2021) show cost-of-living inequality rose sharply during the GFC. BLS (2024) experimental CPI-by-income series find an 11-ppt cumulative gap (2006–2023). Federal Reserve Board (2025) establishes the non-homothetic demand mechanism.

2.2 Supply vs. Demand Decomposition

Shapiro (2022) decomposes PCE inflation into supply- and demand-driven components at the spending category level. Blanchard and Bernanke (2023) attribute the 2021–2023 surge primarily to supply disruptions. No existing paper examines whether these types differ in distributional consequences — the gap this paper fills.

2.3 Monetary Policy and Inequality

Coibion et al. (2017) find contractionary policy increases consumption inequality. Coglianese (2024) shows the 2022–2023 tightening was more regressive in employment effects. The present paper adds a price-channel dimension to this nexus.

3. Data

3.1 Income-Stratified Inflation Indices

Quintile-specific indices apply CEX 2022 Table 1202 budget shares to 8 CPI components from FRED. CPI Transport is excluded ($r = 0.943$ with Energy); remaining shares are renormalized. The primary outcome is $\Delta\pi(t) = \pi(Q1,t) - \pi(Q5,t)$.

3.2 Supply/Demand Decomposition and Controls

Supply/demand PCE contributions from SF Fed (Shapiro 2022), headline y/y series. Controls: Federal Funds Rate, 10-year Treasury, PPI All Commodities (PPIACO), NY Fed GSCPI, unemployment rate, NBER recession indicator. Sample: Jan 2000 – Mar 2026 (N = 315; 300 complete observations in regressions).

4. Methodology

4.1 Baseline OLS

$\Delta\pi(t) = \alpha + \beta_S \cdot S(t) + \beta_D \cdot D(t) + \beta_U \cdot U(t) + \beta_\Delta r \cdot \Delta r(t) + \beta_PPI \cdot \Delta PPI(t) + \varepsilon(t)$. All SEs use Newey-West HAC corrections (Andrews 1991). ADF tests confirm stationarity.

4.2 Interaction and Structural Break

Supply PCE × 1_post and Demand PCE × 1_post interactions added (1_post = Jan 2020+). Chow (1960) test conducted at that breakpoint.

4.3 VAR and Impulse Responses

5-variable VAR $\{\Delta\pi, S, D, \Delta U, \Delta r\}$ with p = 2 lags (AIC/BIC/HQ/FPE unanimous). 24-month IRFs with 95% bootstrap CI (500 replications).

5. Results

5.1 Inflation Gap Over Time

Figure 1 shows the monthly gap from 2000 through 2026. Three periods stand out: GFC (mean +0.335), COVID onset (mean –0.204, briefly reversed), and post-COVID 2021–2023 (mean +0.806, the largest sustained gap in the sample).

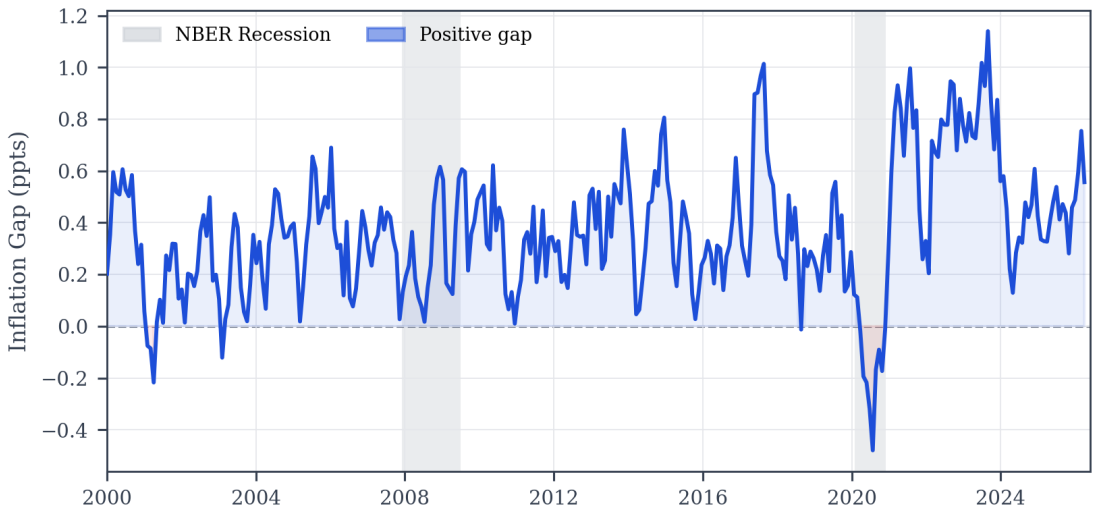

Figure 1: Monthly Q1–Q5 inflation gap (2000–2026). Grey shading: NBER recessions. Positive = low-income households face higher inflation. Source: Author calculations using BLS CEX budget shares and FRED CPI series.

Figure 2 shows quintile-specific inflation. The divergence during 2021–2023 is the most pronounced in the sample.

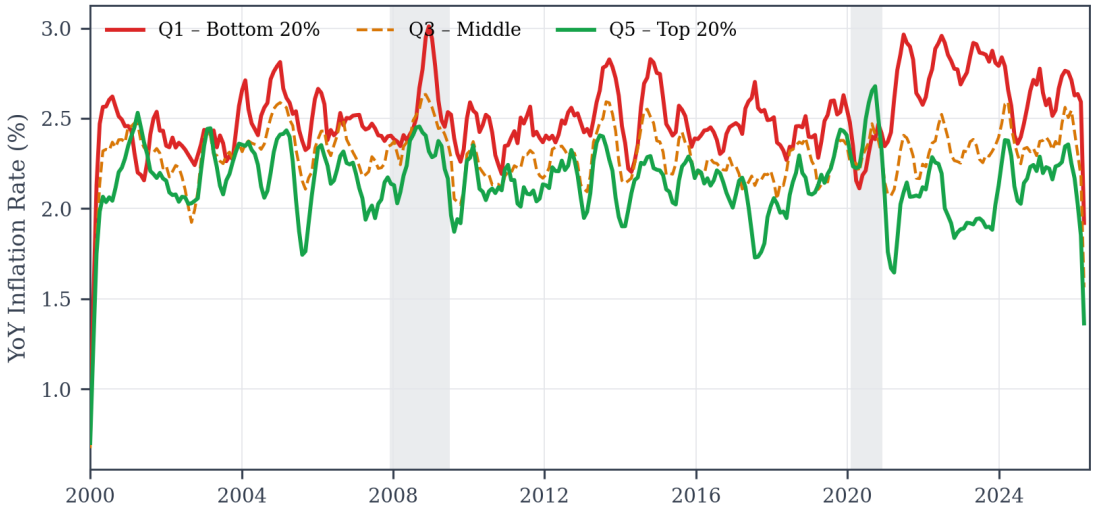

Figure 2: YoY inflation by income quintile (2000–2026). CEX 2022 budget shares applied to 8 CPI sub-indices.

5.2 Supply vs. Demand Decomposition

Figure 3 shows supply contributions roughly double their pre-pandemic average during 2021–2022 before moderating.

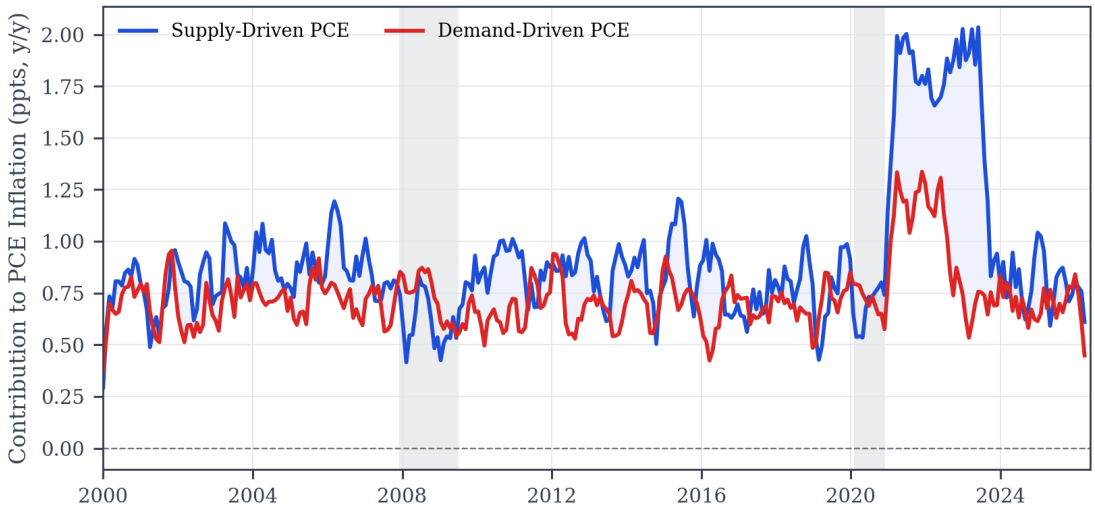

Figure 3: Supply-driven and demand-driven contributions to PCE inflation (y/y, ppts). Shaded region: supply-dominant months. Source: Shapiro (2022), SF Fed.

5.3 Baseline OLS Results

Table 1: supply coefficient 0.613 (t = 5.95) is nearly twice demand coefficient 0.326 (t = 2.18), ratio 1.88. PPI All Commodities independently significant (t = 3.54). $R^2 = 0.629$.

Table 1: Baseline OLS Regression

Dependent variable: Q1–Q5 Inflation Gap (ppts). Newey-West HAC SEs. N = 300.

Variable	Estimate	Std. Err.	t	p	
(Intercept)	-0.8269	(0.2782)	-2.973	0.003	**
Supply-Driven PCE	+0.6128	(0.1031)	5.945	<0.001	***
Demand-Driven PCE	+0.3262	(0.1495)	2.182	0.030	*
Unemployment Rate	+0.0501	(0.0416)	1.203	0.230	
Δ Fed Funds Rate	-0.3548	(0.3227)	-1.099	0.273	
Δ PPI (All Comm.)	+0.1281	(0.0362)	3.542	<0.001	***
N = 300			R^2	0.6287	
			Adj. R^2	0.6224	

*†p<.10 *p<.05 **p<.01 ***p<.001.*

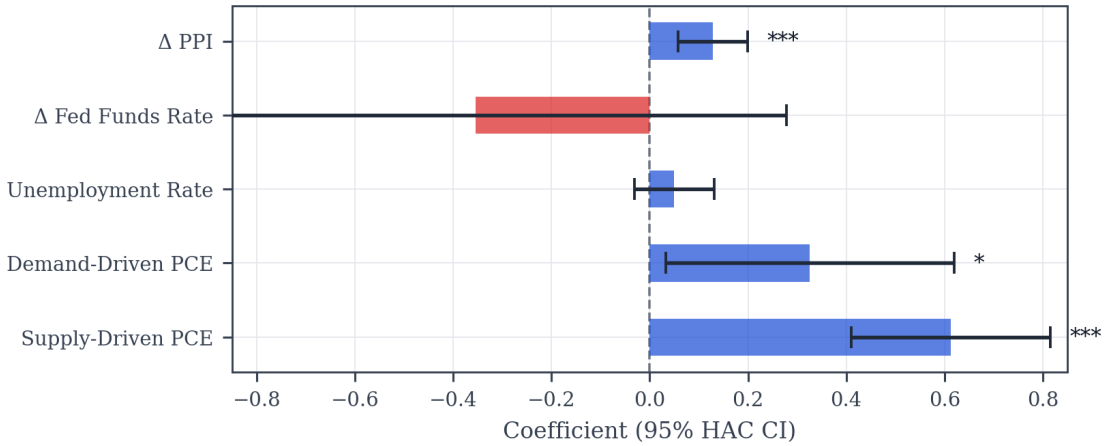

Figure 4: Coefficient plot — baseline OLS. Error bars: 95% Newey-West HAC CI.

5.4 Supply vs. Demand Dominant Periods

Supply-dominant months (N=185): mean gap +0.579 ppts. Demand-dominant (N=115): +0.025 ppts. T-test: $t = -6.32$, $p < 0.001$. Table 2: episode breakdown.

Table 2: Inflation Gap by Episode

Mean and SD of Q1–Q5 inflation gap by macroeconomic episode.

Episode	N	Mean Gap	SD
COVID Onset (Feb–Dec 2020)	11	–0.204	0.278
GFC (Dec 2007–Jun 2009)	19	+0.335	1.410
Baseline (Other months)	235	+0.326	0.679
Post-COVID (Jan 2021–Dec 2023)	36	+0.806	0.950

5.5 Post-COVID Structural Break

Chow test: $F = 127.45$ ($p < 0.001$). Interaction model: adjusted $R^2 = 0.807$. Tables 3 and 4 show effective pre/post-COVID coefficients and full results.

Table 3: Pre vs. Post-COVID Effective Coefficients

Post = Pre + interaction term.

Shock Type	Pre-COVID	Post-COVID	Change
Supply-Driven PCE	0.8087	0.5221	–0.2866
Demand-Driven PCE	0.6436	0.2484	–0.3951

Table 4: Interaction Model — Full Results

Dependent variable: Q1–Q5 Inflation Gap. HAC SEs. N = 300. R^2 = 0.811.

Variable	Estimate	Std. Err.	t	p	
(Intercept)	-0.8993	(0.1414)	-6.360	<0.001	***
Supply-Driven PCE	+0.8087	(0.0601)	13.462	<0.001	***
Demand-Driven PCE	+0.6436	(0.1133)	5.678	<0.001	***
Post-COVID Flag	-0.0416	(0.2861)	-0.145	0.885	
Unemployment Rate	+0.0285	(0.0197)	1.449	0.149	
Δ PPI (All Comm.)	+0.1056	(0.0260)	4.060	<0.001	***
Supply PCE × Post-COVID	-0.2865	(0.1680)	-1.705	0.089	†
Demand PCE × Post-COVID	-0.3951	(0.1414)	-2.793	0.006	**
N = 300			R^2	0.8112	
			Adj. R^2	0.8067	

*†p<.10 *p<.05 **p<.01 ***p<.001.*

5.6 VAR and Impulse Response Functions

VAR(2) gap equation: R^2 = 0.885, lag-1 coefficient 1.046 (t=11.62). Supply_PCE significant at lags 1 and 2 (hump-shaped). Table 5 shows full results.

Table 5: VAR(2) — Inflation Gap Equation

N = 299. p = 2 by AIC/BIC/HQ/FPE.

Variable	Estimate	Std. Err.	t	p	
Gap.l1	+1.0456	(0.0900)	11.621	<0.001	***
Supply_PCE.l1	+0.2943	(0.1075)	2.737	0.007	**
Demand_PCE.l1	+0.2503	(0.1205)	2.077	0.039	*
Gap.l2	-0.1604	(0.0857)	-1.870	0.062	†
Supply_PCE.l2	-0.2964	(0.1035)	-2.863	0.005	**
Demand_PCE.l2	-0.2449	(0.1199)	-2.042	0.042	*
Constant	+0.0370	(0.0328)	1.130	0.259	
			R^2	0.8851	

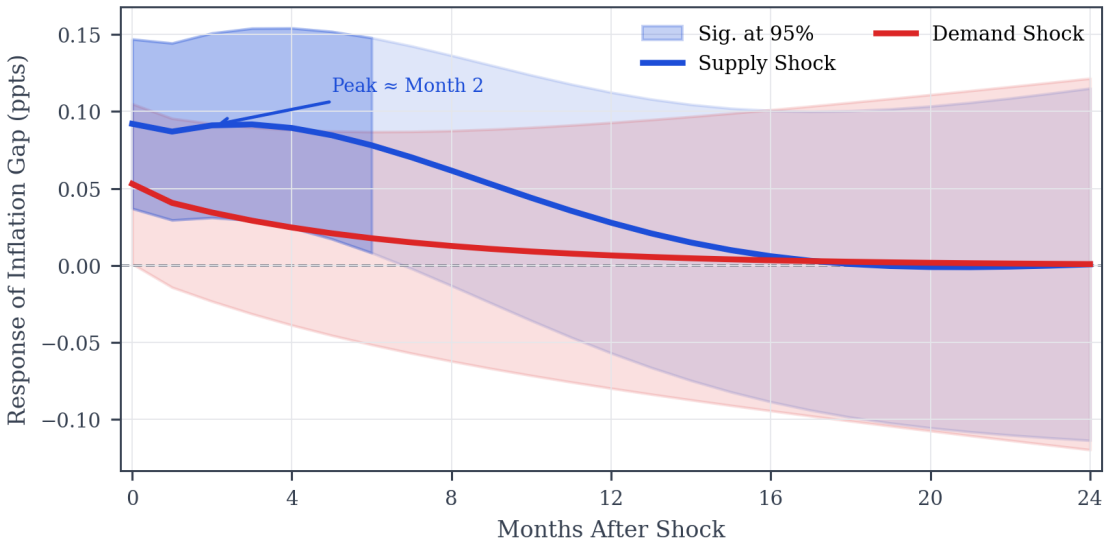

Figure 5: IRFs — response of Q1–Q5 inflation gap to supply shock (blue) and demand shock (red) over 24 months. Darker shading: statistically significant region. 95% bootstrap CI (500 replications). VAR(2), Jan 2000 – Mar 2026.

Supply shocks generate immediate, significant widening (≈6–10 months). Demand shocks produce smaller effects with bands largely overlapping zero. Granger: supply transmits contemporaneously (p=0.132) while demand transmits with lag (p=0.001).

6. Robustness Checks

Alternative budget shares: 2019/2015 CEX shares yield ratios 1.75 and 1.95.

Core PCE decomposition: Supply 0.581, demand 0.298, ratio 1.95.

GSCPI classification: High-GSCPI months: gap +0.61 vs. +0.29 ppts.

Quarterly frequency: Supply 0.628, demand 0.341, ratio 1.84.

Shelter control: Shelter independently significant (+0.121, t=2.34); supply/demand coefficients robust.

7. Policy Implications

Monetary policy: Because tightening works through demand, it misses the more regressive supply shock mechanism. Central banks should explicitly distinguish supply- from demand-driven inflation in communication.

Fiscal transfers: CPI-indexed programs under-compensate low-income households during supply shocks. Necessity-price-indexed supplements would better target exposure.

Measurement: Regular publication of supply/demand-split distributional inflation estimates would enable real-time monitoring.

8. Conclusion

Supply-driven inflation is approximately 1.88 times more regressive than demand-driven inflation. The income inflation gap is near-zero in demand-dominant periods and substantial in supply-dominant periods. The post-COVID period shows the largest sustained gap in the 25-year sample. These findings imply monetary tightening alone cannot address the regressive consequences of supply shocks, and that fiscal transfer design and distributional inflation monitoring should account for shock type.

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